CRYPTOGRAPHY

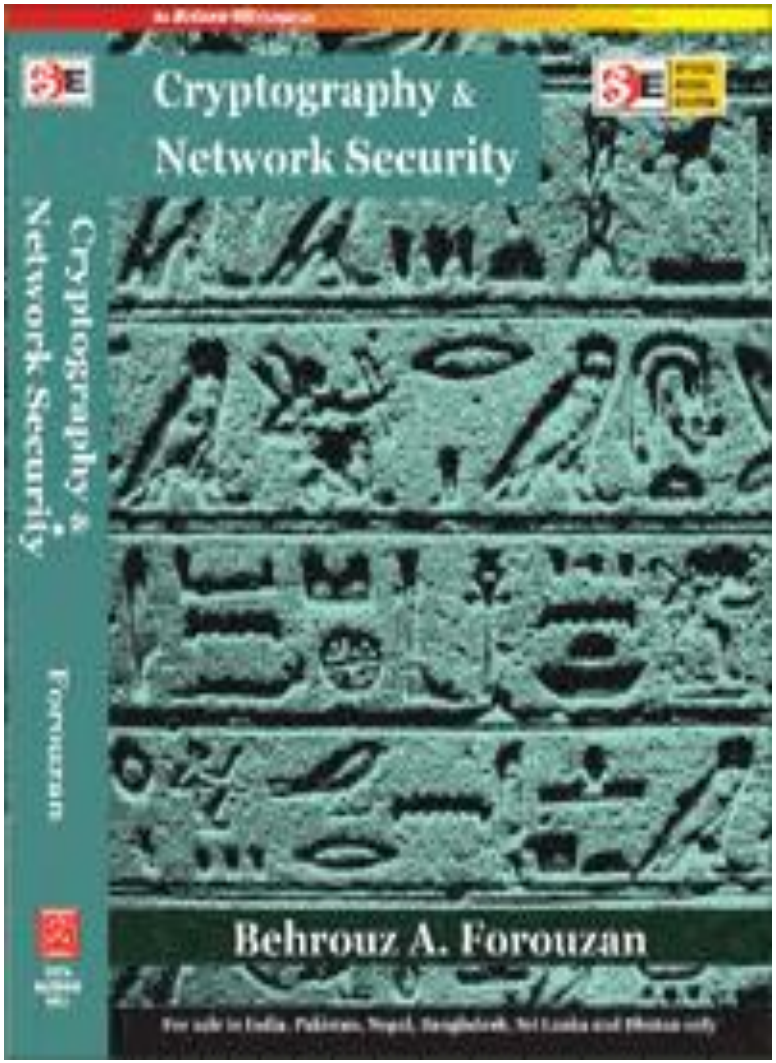
(INFORMATION SECURITY)

- Dr. MANOJ R. MISHRA

- Book:
  - ***Cryptography and Network Security:***

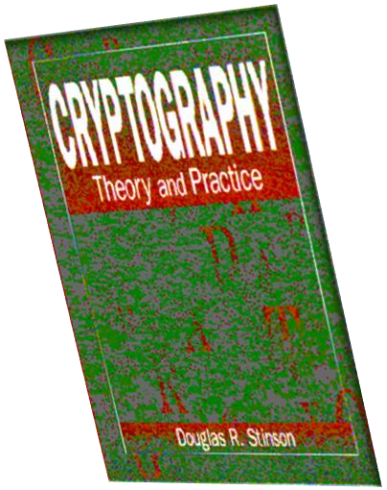
    By **Forouzan B.**

    McGraw-Hill

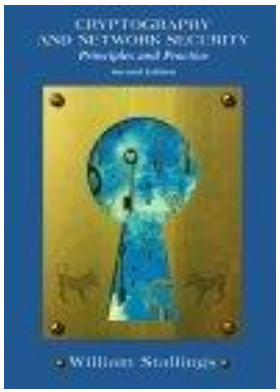


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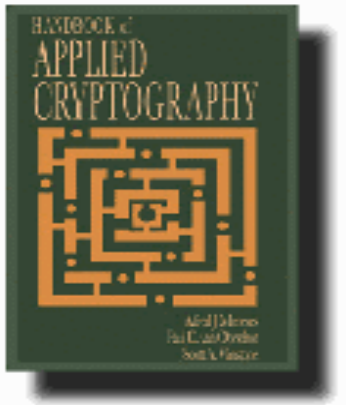
- References:
  - ***Cryptography: Theory and Practice***
    by **Douglas R. Stinson** CRC press



  - ***Cryptography and Network Security: Principles and Practice;***
    By **William Stallings** Prentice Hall



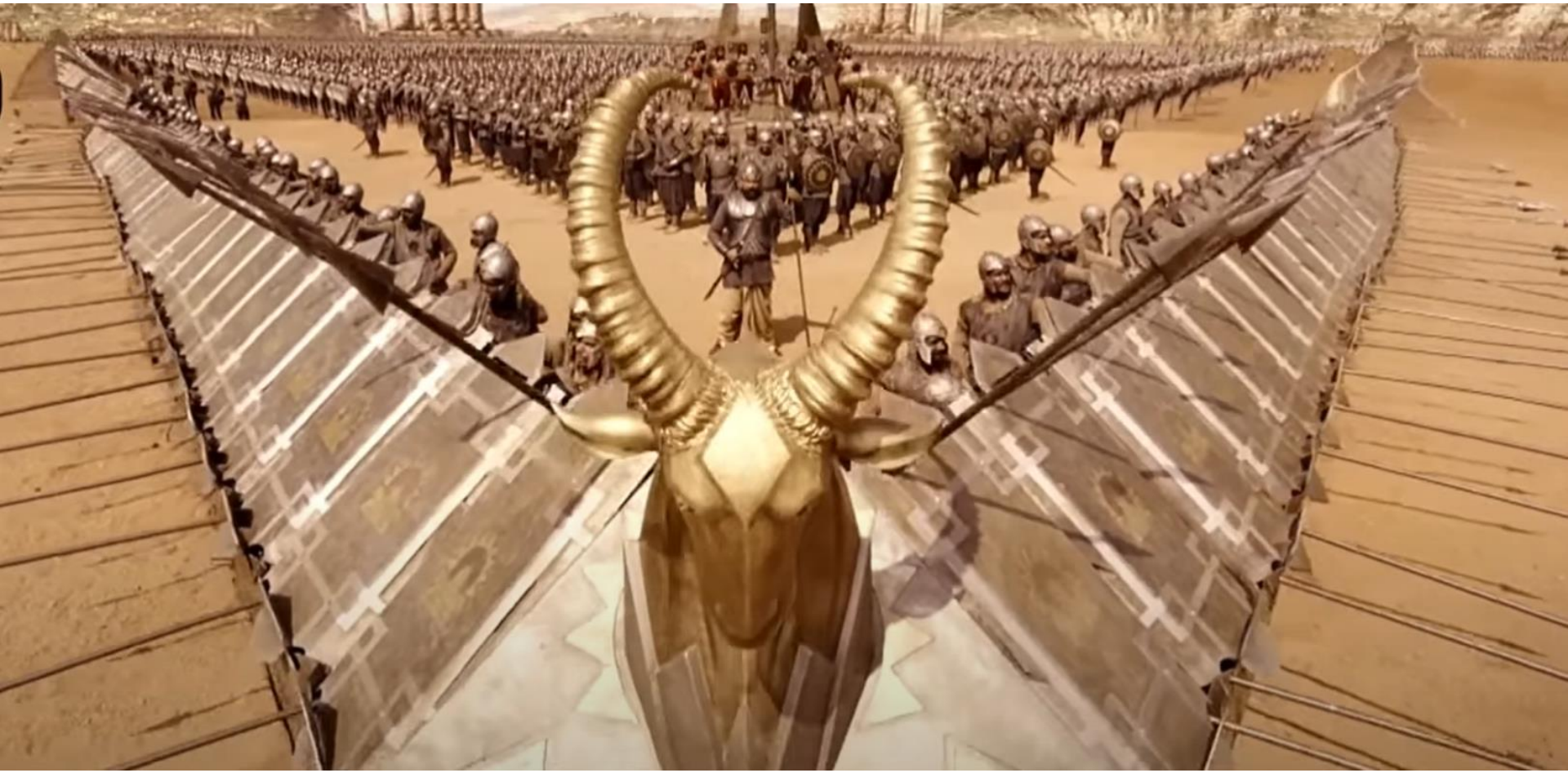
  - ***Handbook of Applied Cryptography***
    by Alfred J. Menezes, Paul C. van Oorschot and Scott A. Vanstone,
    CRC Press



The Art of War







1.6



The art of war teaches us not on the likelihood of the enemy's not coming, but on our own readiness to receive him;

not on the chance of his not attacking, but rather on the fact that we have made our position unassailable.

--The art of War, Sun Tzu

Information Security

- Information is an asset
  - ***Physical file vs. E-file***
  - *From whom it is to be hidden ?*
  - *Who can change ?*
  - *Who can access ?*
  - Authorised vs. Unauthorised access

Cryptography:

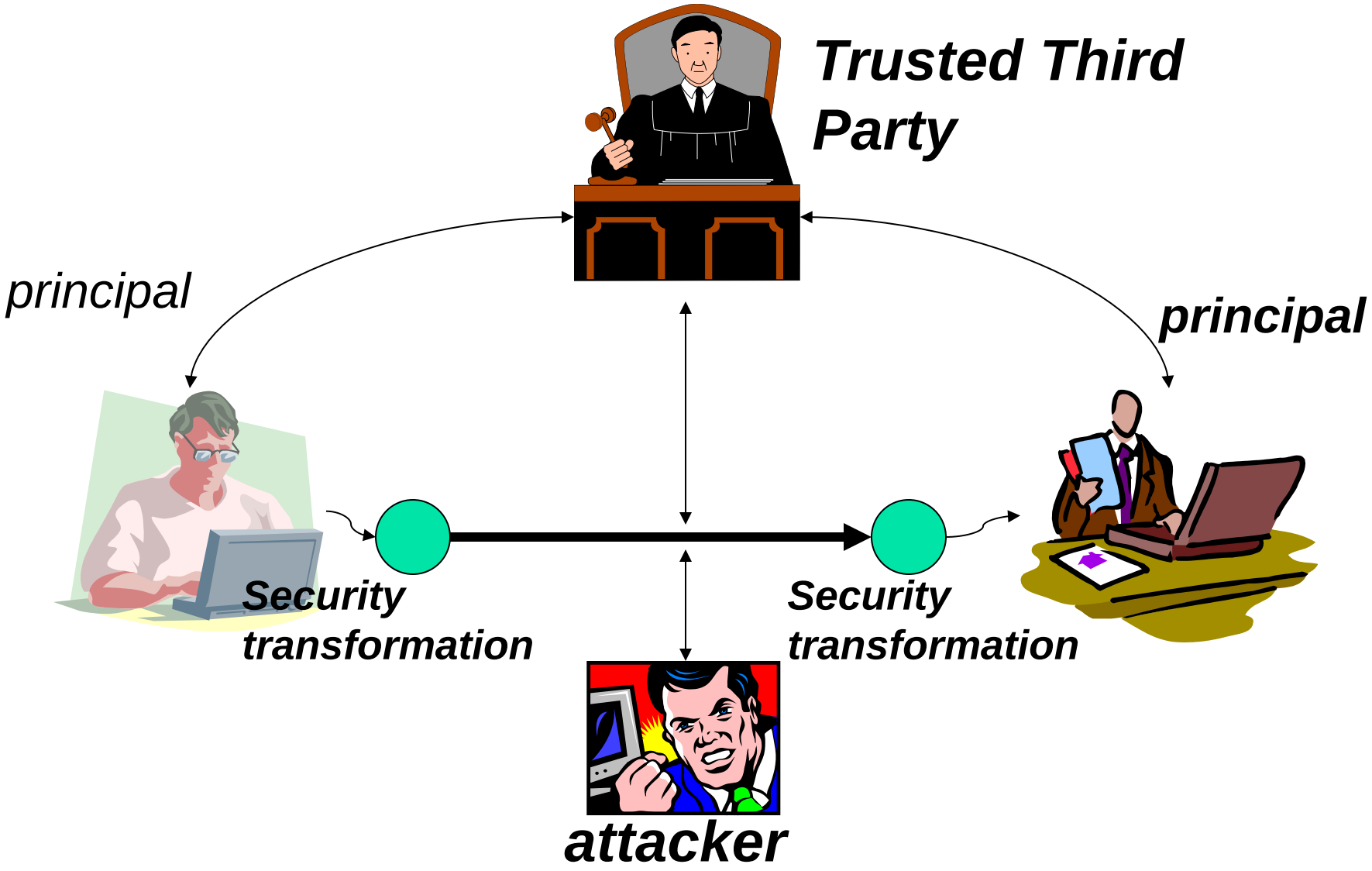
- *from Greek*
  - *kryptós, "hidden",*
  - *and gráphein, "to write"*

- *traditionally, the study of means of converting information from its normal, comprehensible form into an incomprehensible format.*
  - *rendering it unreadable without secret knowledge.*

- *Past: Cryptography helped ensure secrecy in important communications, such as those of spies, military, war, and diplomats.*

- **In recent decades, cryptography has expanded in two ways:**
  - ✓ **mechanisms for more than just keeping secrets:**
  - ✓ **schemes like digital signatures and digital cash, for example.**
  - ✓ **in widespread use by many civilians, and users are not aware about it.**

Network Security Model



Scenario-1

On submitting their assignments online, 4 learners have received their mark sheets and the discussion is as follows:

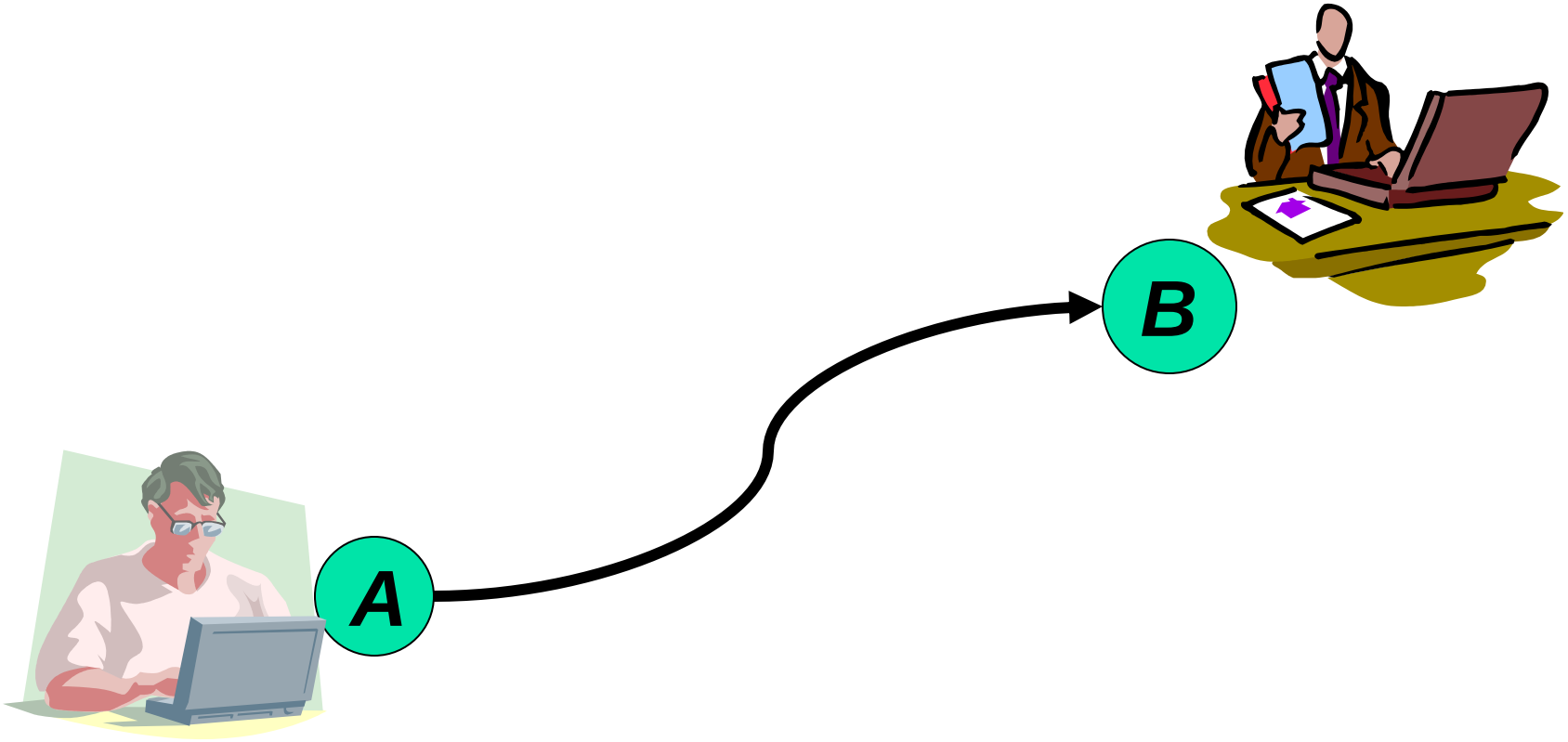
Learner A: ***Oh! my optional subject is SC but I scored 60 in MSS which I have not opted for.***

Learner B: I have answered only two questions yet I secured 80 marks in ICS!

Learner C: I was absent for the ICS exam but to my utter surprise I found 90 marks in it!

Learner D: ***I had done very well in BDA but I have scored 0 in BDA !***

Information Transferring





Define:
SECURITY ATTACK:

Any action that disturbs normal flow of information being sent between sender and receiver is termed as a security attack.

So Information Security is about how

to prevent attacks,

to detect attacks,

on information-based systems.

Kinds of Attack:

Interruption

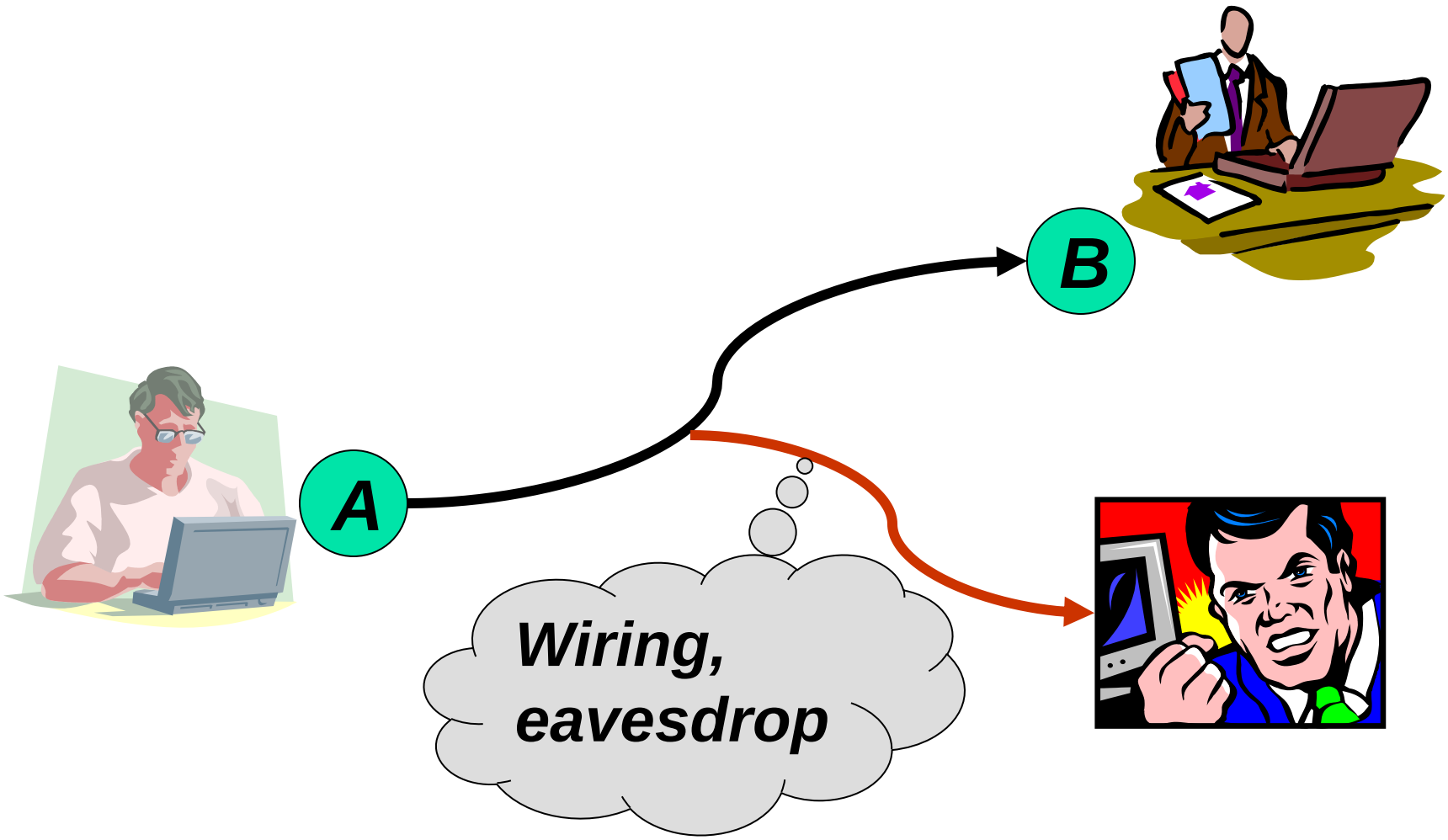




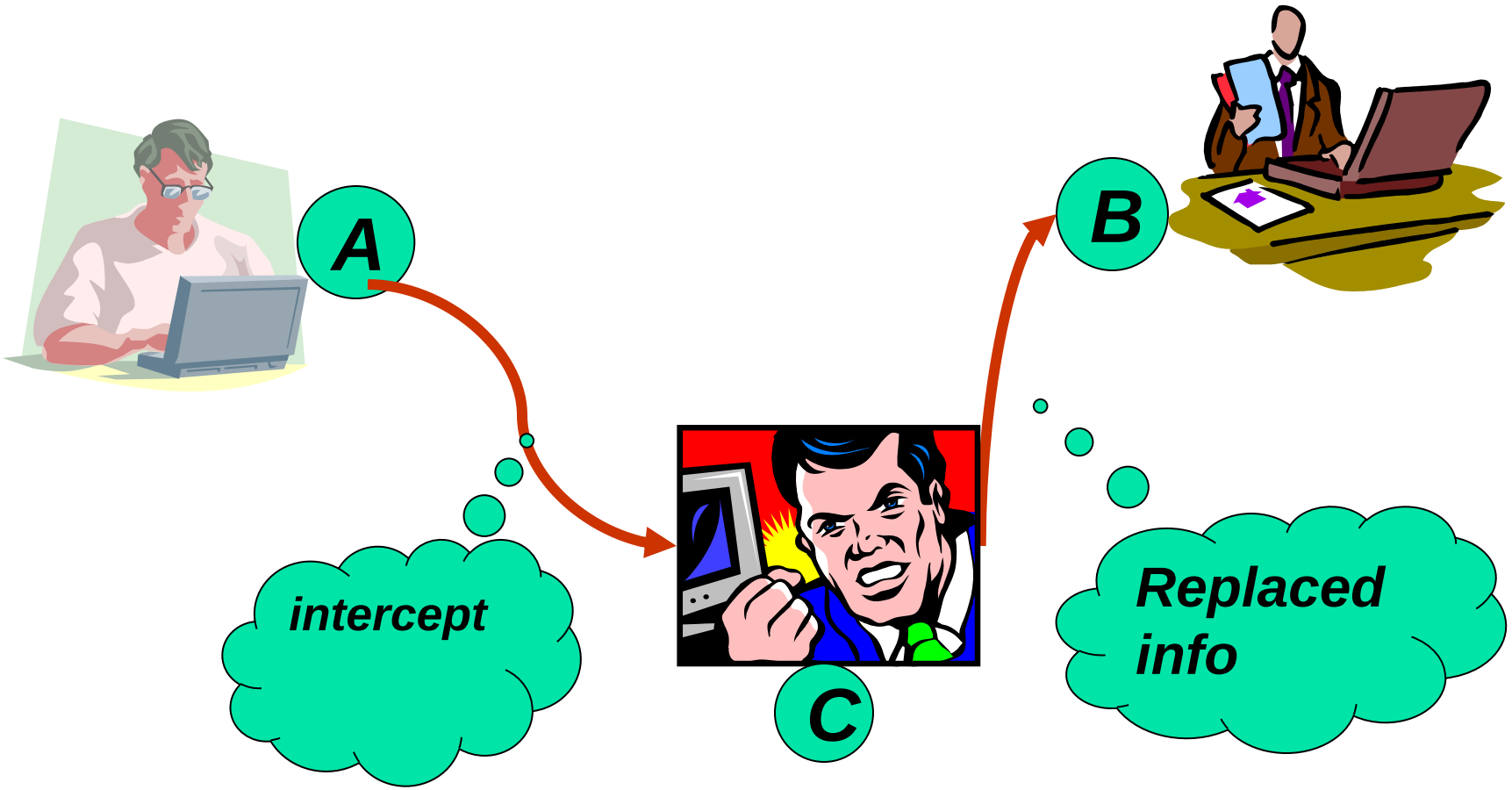
A

Cut wire lines, Jam wireless signals, Drop packets

Attack: Interception



Attack: Modification



Attack: Fabrication







B

C

Also called impersonation

Scenario-2

***Father**: Good morning my boy. After a long Vacation it is time to join the first day of your college and I have credited **Rs 20,000** online in your bank account for your monthly expenditure.*

***Son**: Thank you father, I shall check my account once I reach my hostel.*

Scenario-2 (Contd…)

The series of activities that take place are as follows:

- ***Next day son finds that the amount has not been credited***

  - this depicts attack on availability.

- ***Son e-mails his father and confirms and also requests him to send an additional amount Rs 10,000.***
- ***This information is observed by an intruder and captures the data from a remote location***

  - this depicts attack on confidentiality.

Scenario-2 (Contd…)

- *Next day the father receives sons e-mail and finds that son is requesting him to send Rs 1,00000 more which made him surprised.*
  - *this depicts modification of message*

- *Father confirms once more whether it is the e-mail sent by his son by reconfirming his email ID.*

  But no doubt the senders address was confirmed and he could find that it has come from his son.
  - *this depicts fabrication of data.*

Puzzle:

1) As SERUTAERC is to creatures
so is ENOHPELET is to _____?

Ans:



Puzzle:

2) If word **PEREGRINATION** stands for **1232435678596**, put down the code for the following words:

a) Ration

b) Grip

Ans:

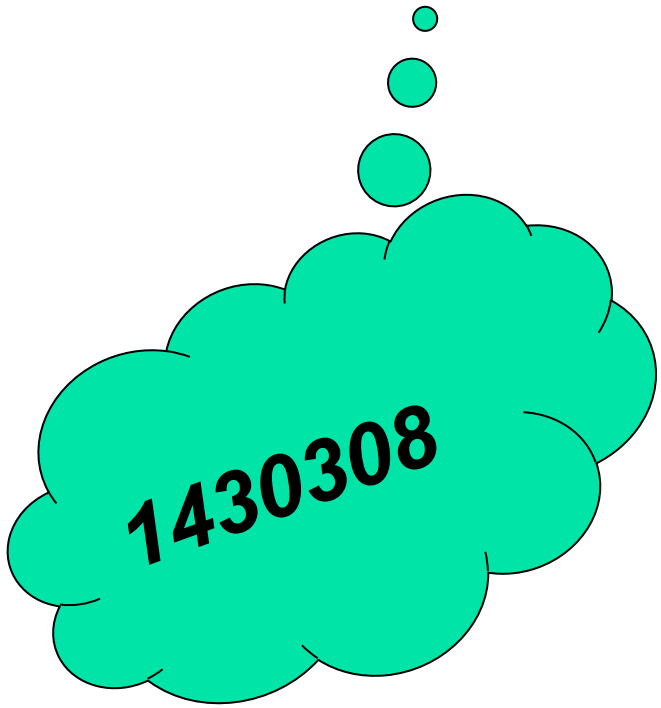




Puzzle:

3)As BAD is to 428, GOOD is to --------?

Ans



Puzzle:

4) As **meet** is to **PHHW**, **party** is to _____?

Ans: SDUWB

Guess Who ?



Julius Caesar was a renowned general, politician and scholar in ancient Rome

Online transactions now require an additional layer of security

The Reserve Bank of India (RBI) has mandated that from 1st Aug 2009 ***all online credit card transactions require an extra level of verification***. Your credit card issuing banks are therefore, implementing a "Verified by VISA" or "MasterCard SecureCode" password.

Apart from the three digit CVV, You will need this password to continue transacting online from 1st Aug 2009.

Verified by VISA

MasterCard SecureCode



How it works?

Enter your credit card details on the payment page → You are redirected to the issuing-bank's website → Enter Verified by VISA / MasterCard SecureCode password → Your transaction is complete



- By National Payments Corporation of India (NPCI)
- All merchant discount rate (MDR) charges were eliminated
- **Unified Payments Interface** (**UPI**) developed by NPCI
- RuPay debit cards are widely accepted within India, Visa debit cards have a higher level of acceptance and can be used at almost all merchants, including international ones.

Q: What is the "Verified by VISA"/"MasterCard SecureCode" password?

A: This is the password that your credit card issuing Bank provides. For example - ICICI Bank, HDFC Bank, Citibank, HSBC Bank, Axis Bank, SBI etc. This password is the additional layer of security for all your online transactions.

Benefits of "Verified by VISA"/"MasterCard SecureCode" password:

Even if you lose / misplace your credit card it cannot be misused online as the password is not present on the card

Remember:

Keep your password safe.

Do not share your cvv/password/OTP with anyone.

SECURITY GOALS (CIA)

This section defines three security goals.

1.1.1 Confidentiality
1.1.2 Integrity
1.1.3 Availability

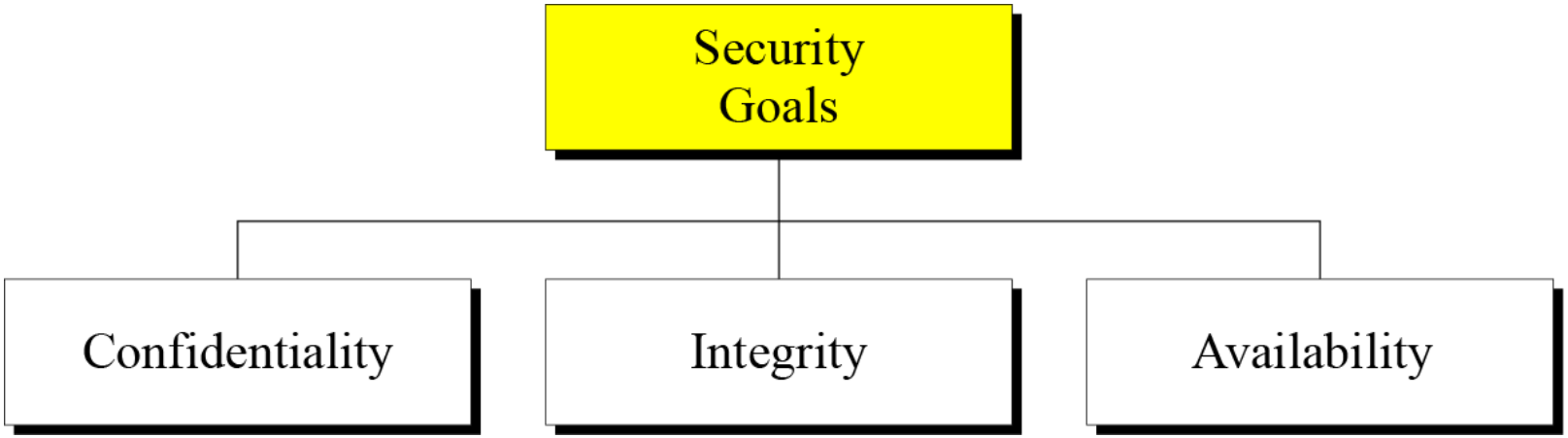


Figure 1.1 *Taxonomy of security goals*

1.1.1 Confidentiality

Confidentiality is probably the most common aspect of information security. We need to protect our confidential information.

An organization needs to guard against those malicious actions that endanger the confidentiality of its information.

B'cos 'INFORMATION IS POWER'

i.e. hide information from unauthorized access

1.1.2 Integrity

- *Information needs to be changed constantly.*

- *Integrity means that changes need to be done only by authorized entities and through authorized mechanisms.*

i.e. Protect from unauthorized change

Ex:BANK- Deposit, Withdraws

1.1.3 Availability

- ***The information created and stored by an organization needs to be available to authorized entities, when it is needed .***

- ***Also Information needs to be constantly changed***

***i.e. At the same time it must be accessible to authorized entities*.**

Attacks Vs. Threates

Threat: ***refers to a potential for violation of security.***
i.e. a potential danger.

Attack: ***an assault on system security.***
i.e. ***an attempt.***

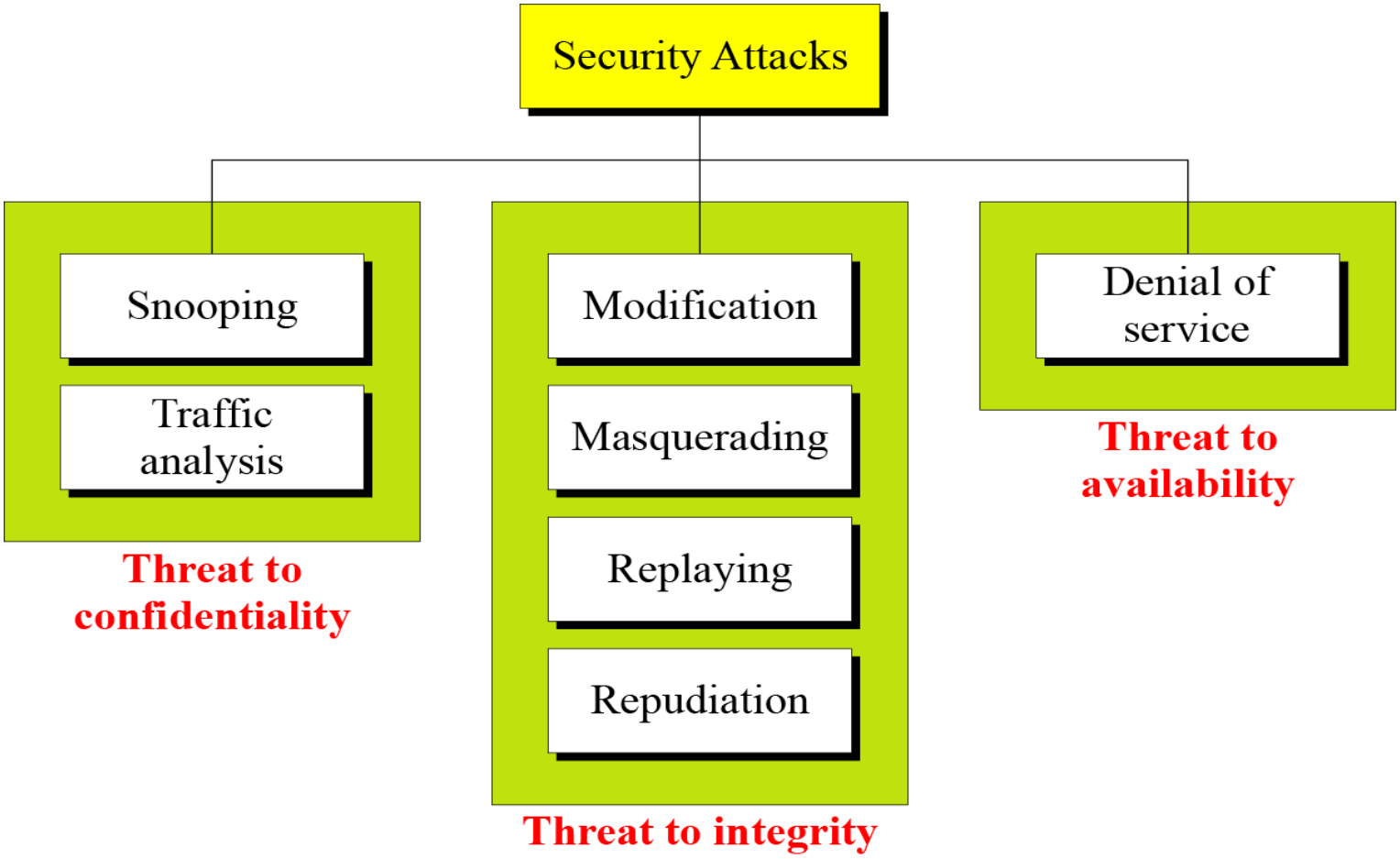
1-2 ATTACKS

The three goals of security - confidentiality, integrity, and availability(CIA) – can be threatened by security attacks.

Topics discussed in this section:

1.2.1 Attacks Threatening Confidentiality
1.2.2 Attacks Threatening Integrity
1.2.3 Attacks Threatening Availability
1.2.4 Passive versus Active Attacks

Figure 1.2 *Taxonomy of attacks with relation to security goals*



1.2.1 Attacks Threatening Confidentiality

Snooping ***refers to unauthorized access to or interception of data.***

Ex: listen to and use it later

Sol: encrypt data

Traffic analysis ***refers to obtaining some other type of information by monitoring online traffic.***

Such as- e-mail id, location/address of sender or receiver, etc.

1.2.2 Attacks Threatening Integrity

Modification ***means that the attacker intercepts the message and changes it.***

Masquerading ***or*** ***spoofing*** ***happens when the attacker impersonates somebody else.***

Replaying ***means the attacker obtains a copy of a message sent by a user and later tries to replay it.***

Repudiation ***means that*** ***sender*** ***of the message might later*** ***deny*** ***that she has sent the message; the*** ***receiver*** ***of the message might later*** ***deny*** ***that he has received the message.***

1.2.3 Attacks Threatening Availability

Denial of service ***(DoS) is a very common attack. It may slow down or totally interrupt the service of a system.***

Vulnerabilities (a weakness or lacuna)

- Human vulnerabilities
  - human behavior or action.
    - Ex: Clicking on a link in an email msg. from a questionable source.
- Protocol vulnerabilities(TCP,IP,ARP,DNS…)
  - The design of security protocols may lead to replay or man-in-the-middle attack.
    - Sniffing passwords from a LAN
- Software vulnerabilities
  - SQL injection, cross-site scripting …
- Configuration vulnerabilities
  - Settings, permissions on files, privileges etc.

1.2.4 Passive vs. Active Attacks

Passive Attacks: ***just to obtain information without affecting the system resoures.***

Active Attacks: ***modify the data stream that affects the resources and their operation.***

Table 1.1 *Categorization of passive and active attacks*

Attacks	*Passive/Active*	*Threatening*
Snooping Traffic analysis	Passive	Confidentiality
Modification Masquerading Replaying Repudiation	Active	Integrity
Denial of service	Active	Availability

Crypto-graphy, -analysis, -logy

- The study of how to circumvent the use of cryptography is called *cryptanalysis*, or *codebreaking*.
- *Cryptography* and *cryptanalysis* are sometimes grouped together under the umbrella term **cryptology**, encompassing the entire subject.
- In practice, "cryptography" is also often used to refer to the field as a whole.
- Cryptography is an interdisciplinary subject,
  - linguistics
  - Mathematics: number theory, information theory, computational complexity, statistics and combinatorics
  - engineering

1-3 SERVICES AND MECHANISMS

ITU-T* provides some security services and some mechanisms to implement those services. Security services and mechanisms are closely related .
because *a mechanism or combination of mechanisms are used to provide a service.*

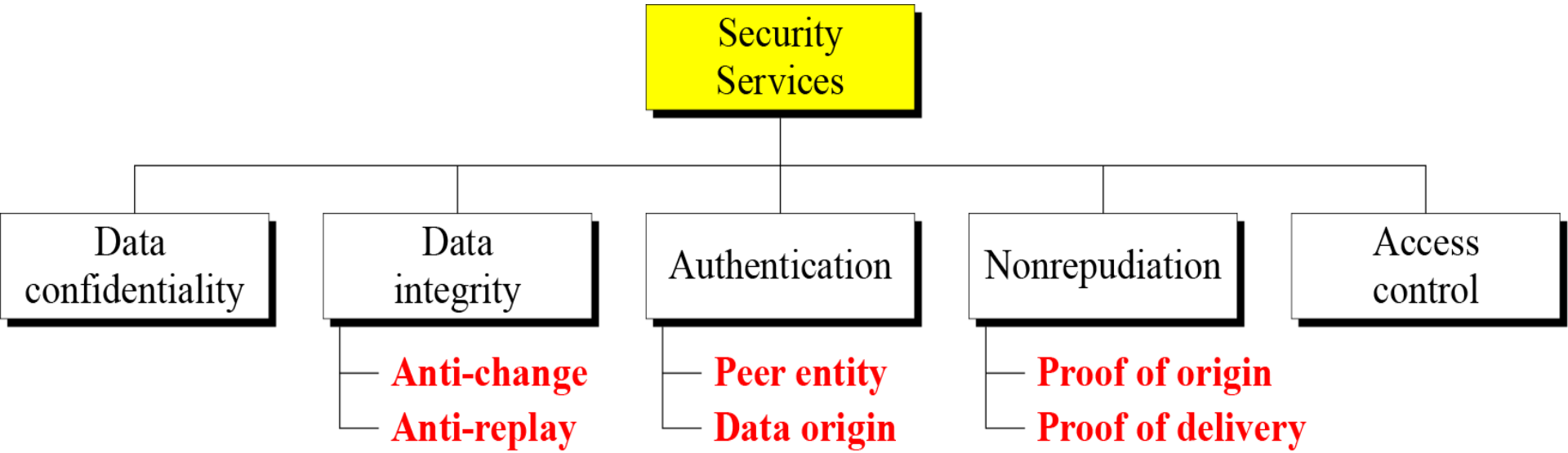
Topics discussed in this section:

1.3.1 Security Services
1.3.2 Security Mechanism
1.3.3 Relation between Services and Mechanisms

**ITU-T: INTERNATIONAL TELECOMMUNICATION UNION –TELECOM. STANDARDIZATION SECTOR*

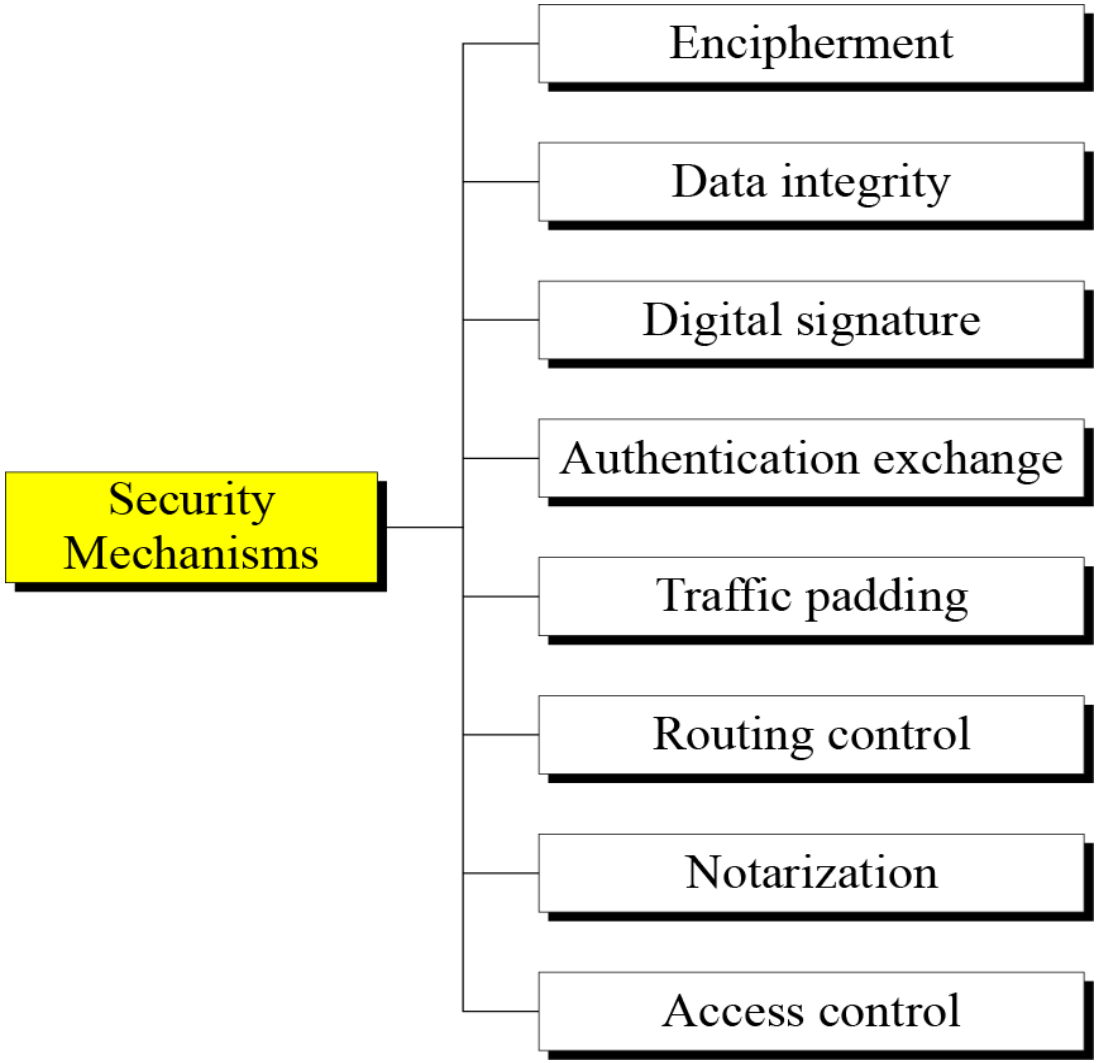
1.3.1 Security Services

Figure 1.3 *Security services*



1.3.2 Security Mechanism

Figure 1.4 *Security mechanisms*



1.3.2 Security Mechanisms

Enciphrement: **hiding** or covering data.

Data integrity: **Sender appends a checkvalue, receiver verifies it.**

Digital Signature: **Sender can electronically sign,**
Receiver verifies it.

Authentication Exchange: **exchange of msg. to prove their identity.**

Traffic padding: **insertion of false data,**
that may prevent traffic analysis

Routing control: **changing different available roots,**
to avoid eavesdropping.

Notarization: **A trusted third-party controls the communication,**
to assure nonrepudiation.

Access control: **access rights to users. Ex: R/W, password, PINs.**

1.3.3 Relation between Services and Mechanisms

Table 1.2 Relation between security services and mechanisms

Security Service	*Security Mechanism*
Data confidentiality	Encipherment and routing control
Data integrity	Encipherment, digital signature, data integrity
Authentication	Encipherment, digital signature, authentication exchanges
Nonrepudiation	Digital signature, data integrity, and notarization
Access control	Access control mechanism

1-4 TECHNIQUES

- *The actual implementation of security goals needs some techniques.*
- *Two techniques are prevalent today:*
  - ***cryptography and steganography.***

Topics discussed in this section:

1.4.1 Cryptography

1.4.2 Steganography

1.4.1 Cryptography

Cryptography, a word with Greek origins, means "secret writing."

However, the term is used to refer to the science and art of transforming messages to make them secure and immune to attacks.

- ***Encryption-Decryption***
  - ***Symmetric-Key Encipherment (secret key)***
  - ***Asymmetric-Key Encipherment (public key-private key)***

1.4.2 Steganography

The word steganography, with origin in Greek, means "covered writing," in contrast with cryptography, which means "secret writing."

Example: covering data with text

This book □ is □□ mostly about □ cryptography, □ not □□ steganography.

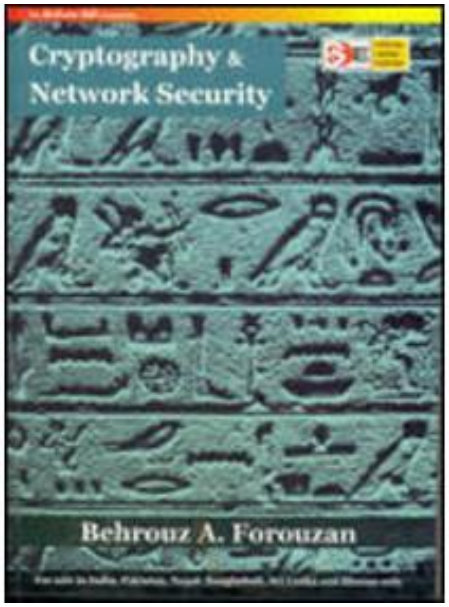
0 1 0 0 0 0 1

A in ASCII - 01000001

Between two words a single space - 0

and a double space - 1

Image Steganography



R + G + B

8 + 8 + 8=24 bits

RGB color model.

Red : RGB(255,0,0) - RGB(11111111, 00000000, 00000000)

Green : RGB(0,255,0) - RGB(00000000, 11111111, 00000000)

Blue : RGB(0,0,255) - RGB(00000000, 00000000, 11111111)

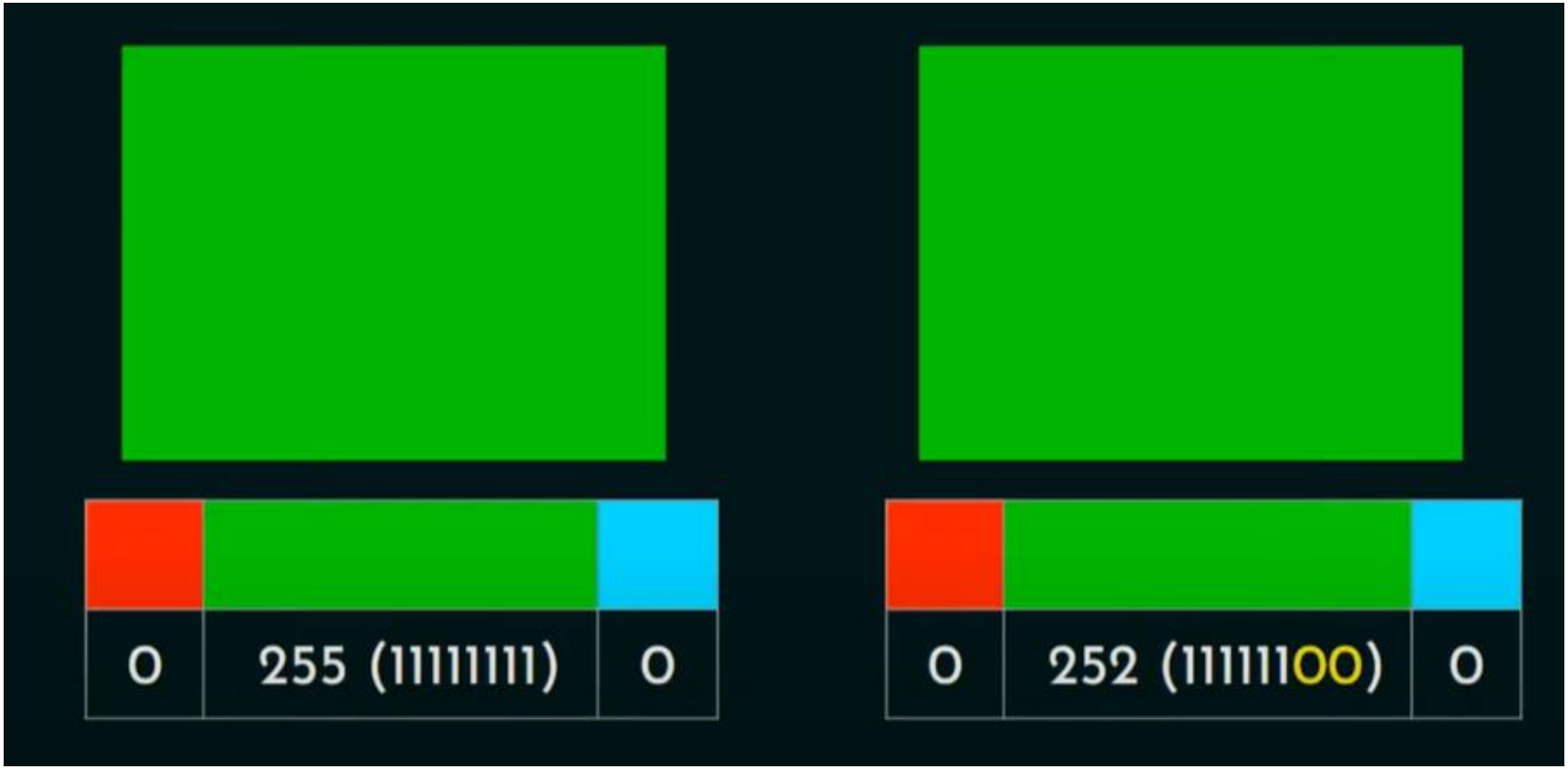
1.4.2 LSB Steganography

Example: covering data under color image

01010011	10111100	01010101
01011110	10111100	01100101
01111110	01001010	00010101



Green: -Find The Difference?



Tools for Stenography

- http://www.jjtc.com/Steganography/toolmatrix.htm

1-5 THE REST OF THE BOOK

The rest of this book is divided into four parts.

Part One: Symmetric-Key Enciphermen

Part Two: Asymmetric-Key Encipherment

Part Three: Integrity, Authentication, and Key Management

Part Four: Network Security